

Problem 1

Line	Count
1	$2n+3$

explain
 $C_i(n/2)$ is 2 ops for $(n/2)+1$
 $1++$ is 2 ops for $n, 2$
 total is $1 + 2((n/2)+1) + 2(n/2) = 2n+3$

Line	Count
2	n

explain
 1 access array 1 assignment
 $= 2 \text{ ops runs } n/2 \text{ times}$
 $2(n/2) = n$

Line	Count
3	$2.5n$

2x PLW
 2 Subtractions 2 access
 1 assignment $= 5 \text{ ops runs } n/2$
 $5(n/2) = 2.5n$

Line	Count
4	$2n$

explain
 2 subtracts 2 access 1 assign
 $= 5 \text{ ops runs } n/2 \text{ times}$
 $5(n/2) = 2.5n$

Line	Count
5	0

explain
 Braces

Summary expression: $7.5n + 3$

Problem 1, c

The notation is $O(n)$

Because the Highest term in the expression $7.5n + 3$ is n we drop the constants and coefficients.
Because we only process half the inputs the time scales linearly.

Bubble sort

$-50, 13, 36, 1, -47, 25, 29, 39, -15$

Pass 1

-50	13	36	1	-47	25	29	39	-15
-50	13	36	1	-47	25	29	39	-15
-50	13	1	36	-47	25	29	39	-15
-50	13	1	-47	36	25	29	39	-15
-50	13	1	-47	25	36	29	39	-15
-50	13	1	-47	25	29	36	39	-15
-50	13	1	-47	25	29	36	39	-15
-50	13	1	-47	25	29	36	-15	39

Bubble Sort

Pass 2

-50	1	1	3	-47	25	29	36	-15	39
-50	1	-47	13	25	29	36	-15	39	
-50	1	-47	13	26	29	-15	36	39	

Pass 3

-50	-47	1	1	3	25	29	-15	36	39
-50	-47	1	13	26	-15	29	36	39	

Pass 4

-50, -47, 1, 13, -15, 25, 29, 36, 39

Pass 5

-50, -47, 1, -15, 13, 25, 29, 36, 39

Pass 6

-50, -47, -15, 1, 13, 25, 29, 36, 39

Selection Sort

-50 13 36 1 -47 25 29 39 -15

Find max and move to end (-15 → 39)

-50 13 36 1 -47 25 29 -15 | 39

Find and swap (36, -15)

-50 13 -15 1 -47 25 29 | 36 39

Find and swap 29 and 25 swap with 25

-50 13 -15 1 -47 | 25 29 36 39

Find and swap (13 -47)

-50 -47 -15 1 13 25 29 36 39

Merge sort

$[-50, 13, 36, 1, -47, 25, 29, 39, -15]$

$[-50, 13, 36, 1, -47]$

$[25, 29, 39, -15]$

$[-50, 13, 36]$

$[1, -47]$

$[25, 29]$

$[39, -15]$

$[-50, 13]$

$[36]$

$[1]$

$[-47]$

$[25]$

$[29]$

$[39]$

$[-15]$

$[-50]$

$[13]$

$[-50, 13]$

$[36]$

$[-47, 1]$

$[25, 29]$

$[-15, 39]$

$[-50, 13, 36]$

$[-47, 1]$

$[-15, 25, 29, 39]$

$[-50, -47, 1, 13, 36]$

$[-15, 25, 29, 39]$

$[-50, -47, -15, 1, 13, 25, 29, 36, 39]$